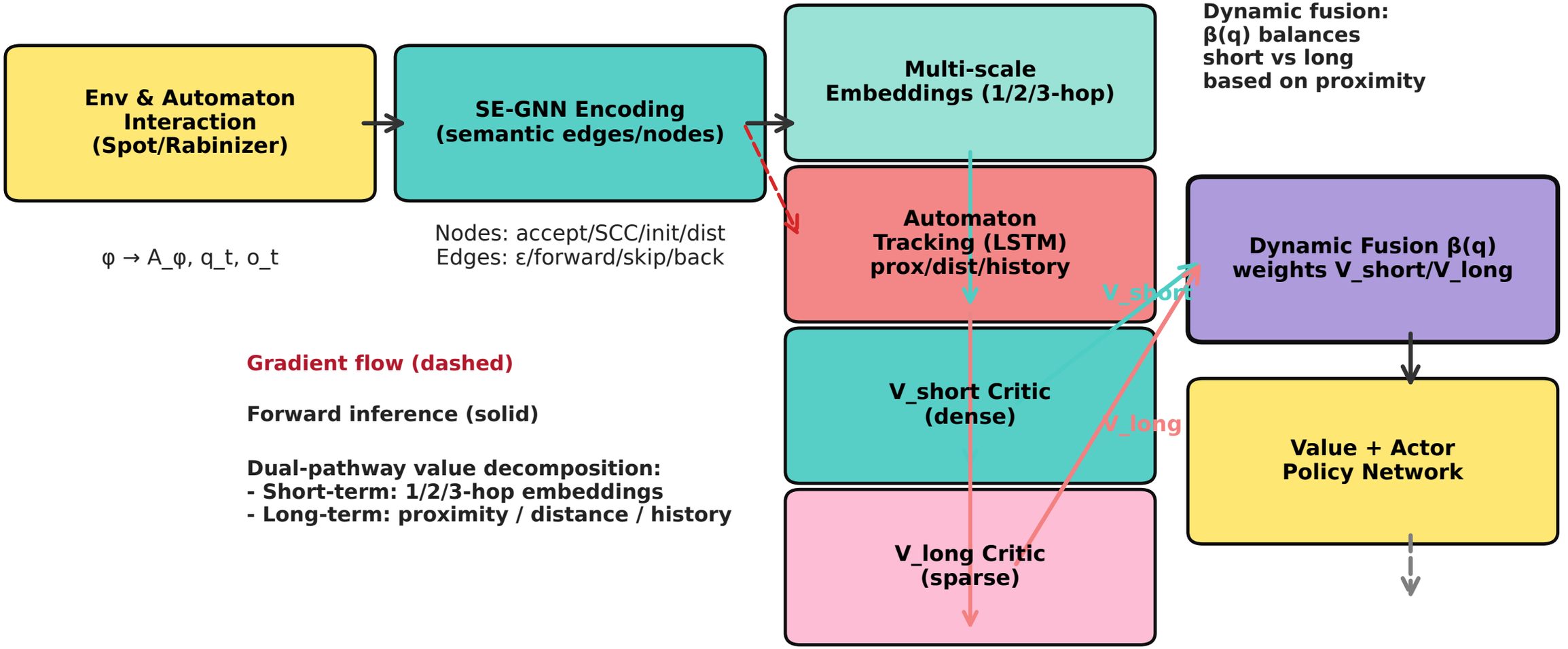
Env & Automaton Interaction (Spot/Rabinizer)
SE-GNN Encoding (semantic edges/nodes)
Multi-scale Embeddings (1/2/3-hop)
Dynamic fusion: β(q) balances short vs long based on proximity
φ → A_φ, q_t, o_t
Nodes: accept/SCC/init/dist
Edges: ε/forward/skip/back
Automaton Tracking (LSTM) prox/dist/history
Dynamic Fusion β(q) weights V_short/V_long
V_short
V_long
V_short Critic (dense)
Value + Actor Policy Network
V_long Critic (sparse)
Gradient flow (dashed)
Forward inference (solid)
Dual-pathway value decomposition:
- Short-term: 1/2/3-hop embeddings
- Long-term: proximity / distance / history